

Gustavo Morales

Senior Data Scientist Astrophysicist



Santiago, Chile
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gus-morales
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About me

Experienced professional with 8 years in academia, plus 5 years in financial analytics and machine learning projects. Proven track record of publications, storytelling, and advanced analytics. Skilled in extracting valuable insights from complex data. Strong problem-solving and communication abilities. Committed to high-quality work and exceeding expectations.

Education

2014 – 2018 | PhD, Astrophysics
Heidelberg University, Germany

2010 – 2013 | MSc, Astrophysics
P. Universidad Católica de Chile

2006 – 2010 | BSc, Astronomy
P. Universidad Católica de Chile

Skills & Tools

- Languages: Python, SQL
- Data manipulation: numpy, pandas, pyspark, polars
- Statistical Modeling: scipy, scikit-learn, statsmodels, xgboost
- Statistical tools and visualization: matplotlib, seaborn, plotly, ggplot2
- Deployment: Airflow, MLflow, Streamlit
- GenAI & LLM Ops: Anthropic Claude API, prompt engineering, LLM-generated reporting
- Data and web scraping: Scrapy, BeautifulSoup
- Cloud microservices: GCP (BigQuery), AWS (EC2, RDS, S3, Lambda, Athena, Glue, Sagemaker, Quicksight)
- Project management: Github, Jira, Confluence

Languages

Spanish, English, German (basic)

Work experience

2024 – 2026 Senior Decision Scientist

Aplazo

At a fintech offering buy-now-pay-later loans across a 15,000+ merchant network, led three machine-learning initiatives end to end, from data engineering to production deployment and monitoring. Delinquency prediction: designed and shipped a two-stage XGBoost hurdle model that predicts delinquent installments per loan at a fixed days-on-book horizon, with probability calibration, empirical-Bayes merchant-level shrinkage, and a daily BigQuery batch-inference pipeline; cut forecast error (MAPE) by 85% versus the previous production model. Built an automated monitoring and evaluation suite (feature-drift PSI, out-of-distribution detection, weighted-MAPE tracking, champion/challenger comparison) that scores predictions against matured actuals and raises degradation alerts. Also built contribution analysis and explanatory tools informing risk and growth decisions.

2024 Data Science Consultant

Contract

RMES Analytics: built production data products for the RMES asset-performance suite, delivering generative-AI features, optimization routines, and ML classification models for predictive maintenance that flag equipment failures before they occur. Smart Mining: architected two end-to-end codebases for a mining operator, taking descriptive, prescriptive, and predictive solutions for mineral comminution and mill maintenance from raw-data ingestion through modeling to deployment.

2020 – 2023 Senior Data Scientist

Banco de Crédito e Inversiones

Delivered corporate-finance analytics and macroeconomic forecasting (IMACEC, PIB) for the Growth division of an online payments platform, using big-data pipelines on AWS. Modeled user behavior and engagement, and built a feature store powering production models with automated monitoring, data-quality checks, and interpretability. Work spanned time-series forecasting, binary/multiclass classification, regression, client segmentation, NLP, and data-driven storytelling.

2019 – 2020 Junior Data Scientist

djavu

Engaged in software engineering projects, leveraging scalable architectures to develop innovative digital solutions. Gained valuable experience at the enterprise level within an agile framework, focusing on workflow management for data engineering pipelines. Played a key role in building, testing, and deploying containerized applications to ensure efficient and scalable solutions.

2014 – 2018 PhD Thesis

Heidelberg University

Conducted extensive analysis by comparing observational data with simulations of stellar streams in external galaxies. Explored the intricate interplay of numerous variables within a rigorous statistical framework, providing detailed insights into their effects. Effectively communicated findings in many technical talks, university visits and symposia, including the discovery of numerous previously unpublished astrophysical objects.

2010 – 2013 MSc Thesis

P. Universidad Católica

Characterized and exploited data maps generated by a major radiotelescope via computation of relevant physical quantities, noise distribution and so on, for a variety of bright sources. The algorithm involved the calculation of an optimal filter to maximize data quality and highlight any particular features of interest.

2006 – 2010 BSc Thesis

P. Universidad Católica

Performed a comprehensive analysis of a specific cosmological simulation, employing rigorous statistical tests to evaluate hypothetical observations related to superclusters of galaxies.

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References

Chris Bucknell
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Business Analytics Team Lead,
Aplazo

Dr. Rodrigo Rubio
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Data & Analytics Team Lead,
Banco de Crédito e Inversiones

Dr. Fabián Reyes
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Investment Analytics Team Lead,
Banco de Crédito e Inversiones

Daniel Aviv
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CEO, Soluciones Digitales djavu

Prof. Dr. Eva K. Grebel
grebel@ari.uni-heidelberg.de
Director Astronomisches
Rechen-Institut Zentrum für
Astronomie der Universität
Heidelberg

Freelance Projects

2009 Pontificia Universidad Católica
Used statistical analysis to further develop a newly found correlation between the initial mass function, metallicity and the presence of some elements for a set of particular star clusters.

2008 Cerro Tololo Interamerican Observatory
Cross-matched catalogs of galaxy clusters to derive new correlations and statistical descriptions.

2006 Pontificia Universidad Católica
Worked in a bigravity theory that naturally predicts dark matter and successfully reproduces the expansion history of the universe.

Teaching Experience

Astronomical Techniques,
Prof. E. Grebel, Heidelberg University

Orders of Magnitude,
Prof. A. Reisenegger
Pontificia Universidad Católica de Chile

Thermodynamics
Prof. S. Hevia,
Pontificia Universidad Católica de Chile

Classical Physics,
Prof. G. Galaz
Pontificia Universidad Católica de Chile

Contemporary Physics,
Prof. G. Garcia
Pontificia Universidad Católica de Chile

Publications

- Morales, G. 2018, PhD Dissertation, DOI: 10.11588/heidok.00025233
- Morales, G., Martínez, D., Grebel, E. K., et al. 2018, A&A, 614, A143.
- Romanowsky, A. J., Martínez, D., Morales, G. et al. 2016, MNRAS, 457, L103.
- Martínez, D., D'Onghia, E., Morales, G., et al. 2015, AJ, 150, 116.
- Gralla, M. B., Crichton, D., Morales, G., et al. 2014, MNRAS, 445, 460.
- Marsden, D., Gralla, M., Morales, G., et al. 2014, MNRAS, 439, 1556.

Other Education and certifications

2023 Bootcamp: Data Science for the Banking Industry

[x-academy](#)

2019–2020 Bootcamp: Data Science

[Desafío Latam](#)